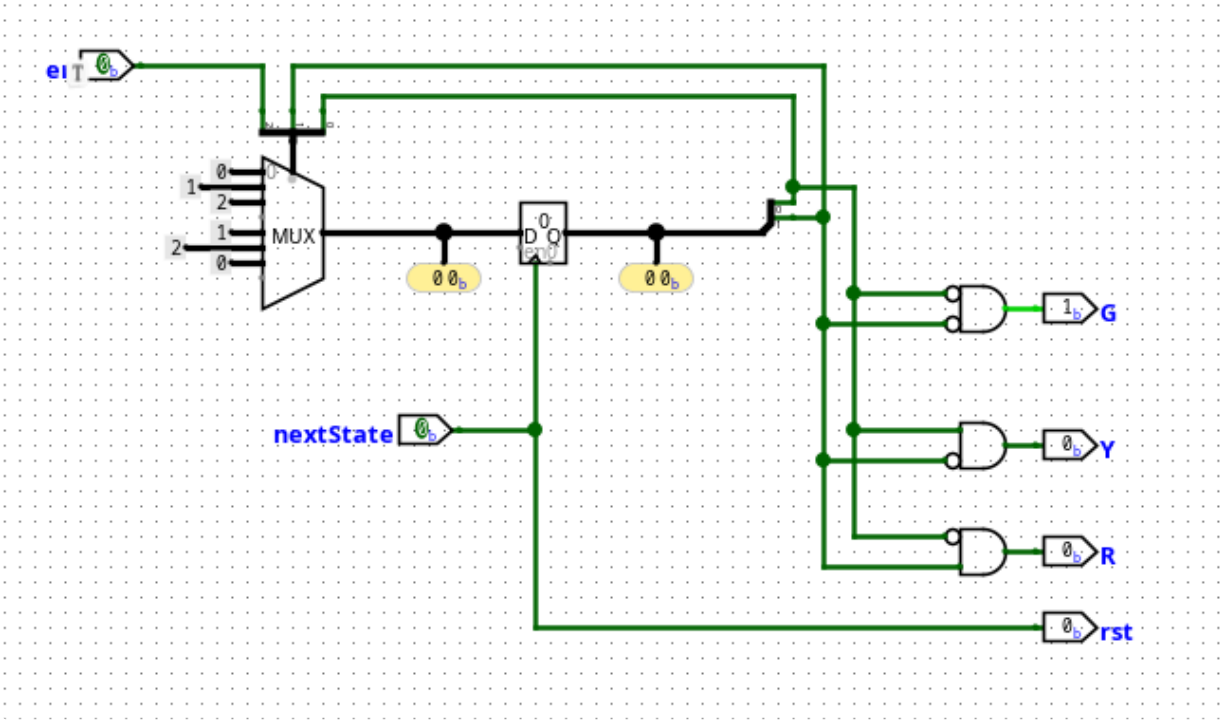
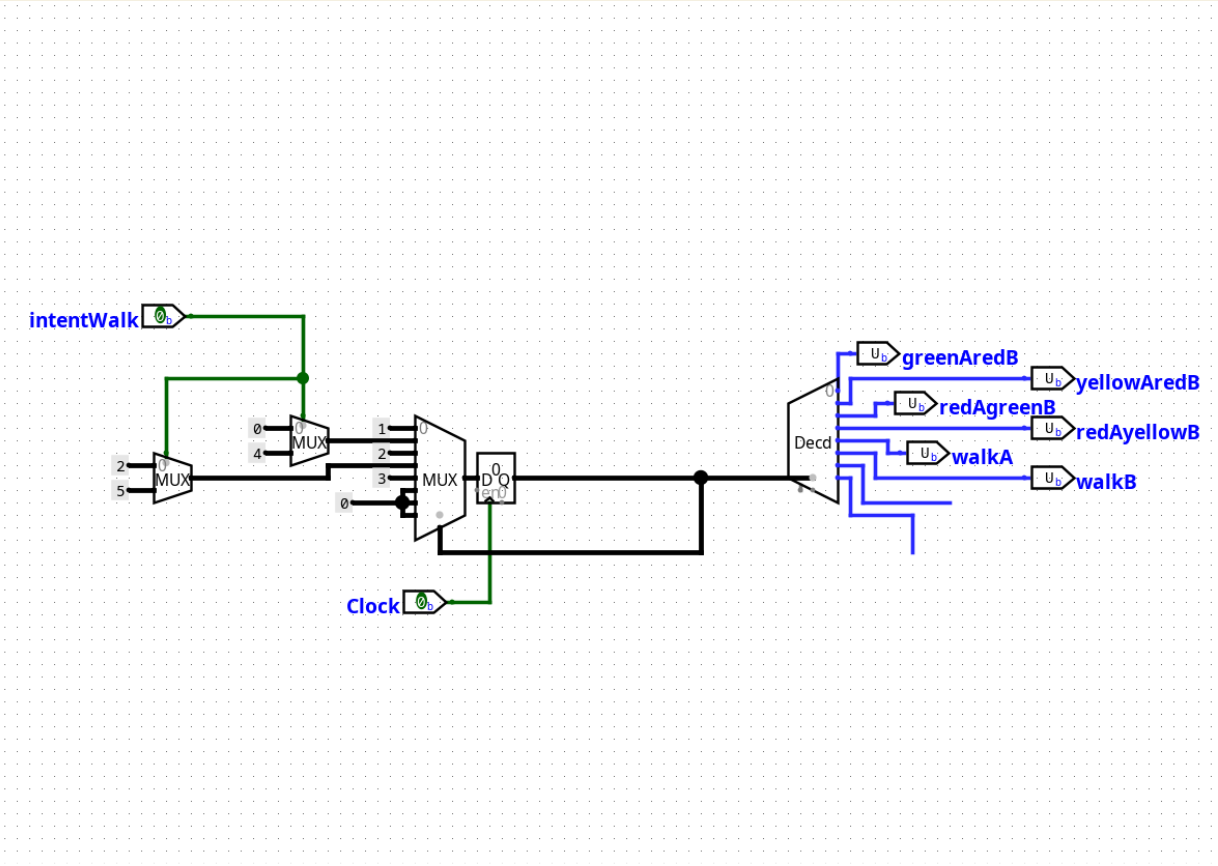
Final Lab (Traffic Lights)

The goal for our final lab is to make a fully functional Traffic Light--pedestrian requests, car cycles, following the same 'trajectory' of Lab 7.

This is the initial FSM we built in Lab 7. However, it only works for one light, so I made my own iteration;



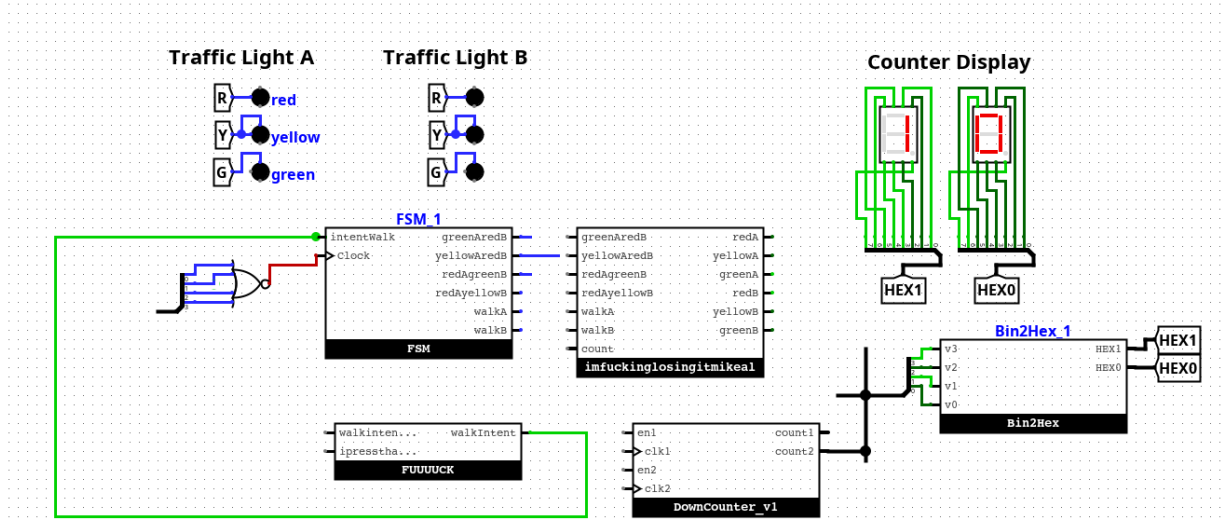
including a multiplexer that has specific toggles for when walkIntent can be activated, connecting to another multiplexer.

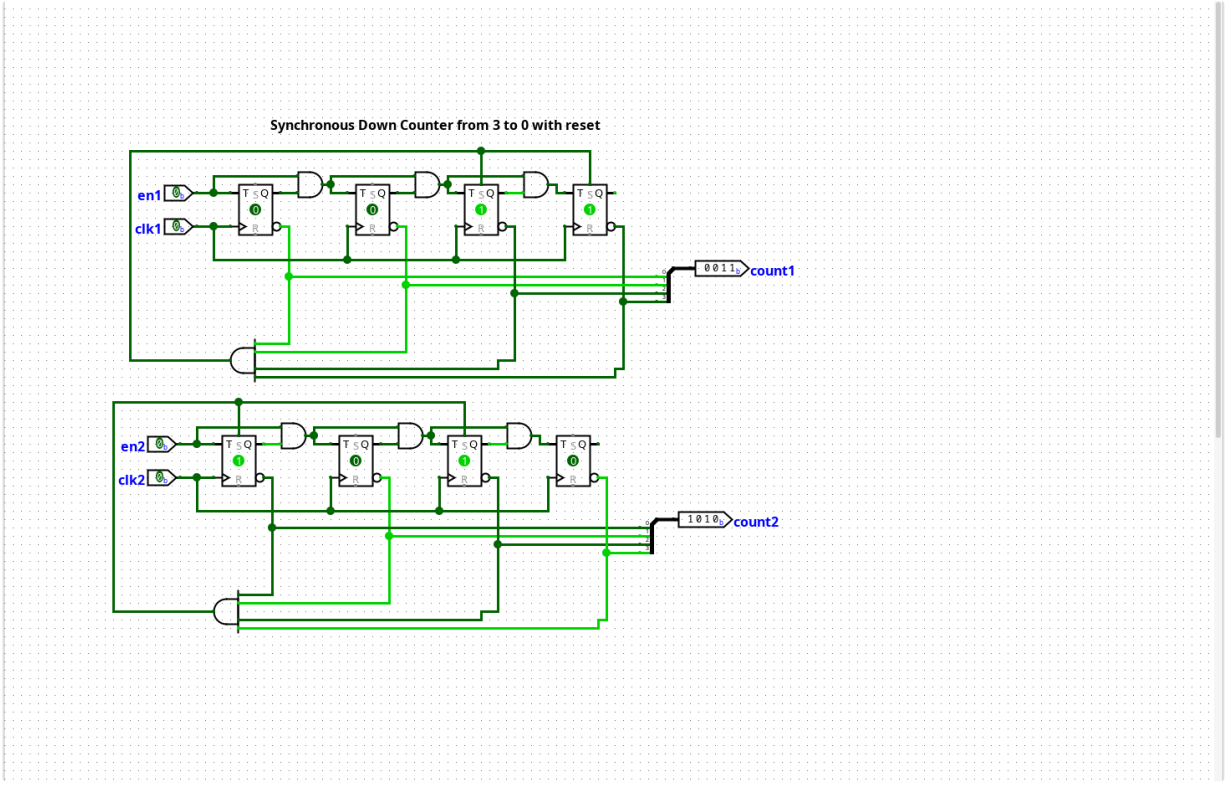


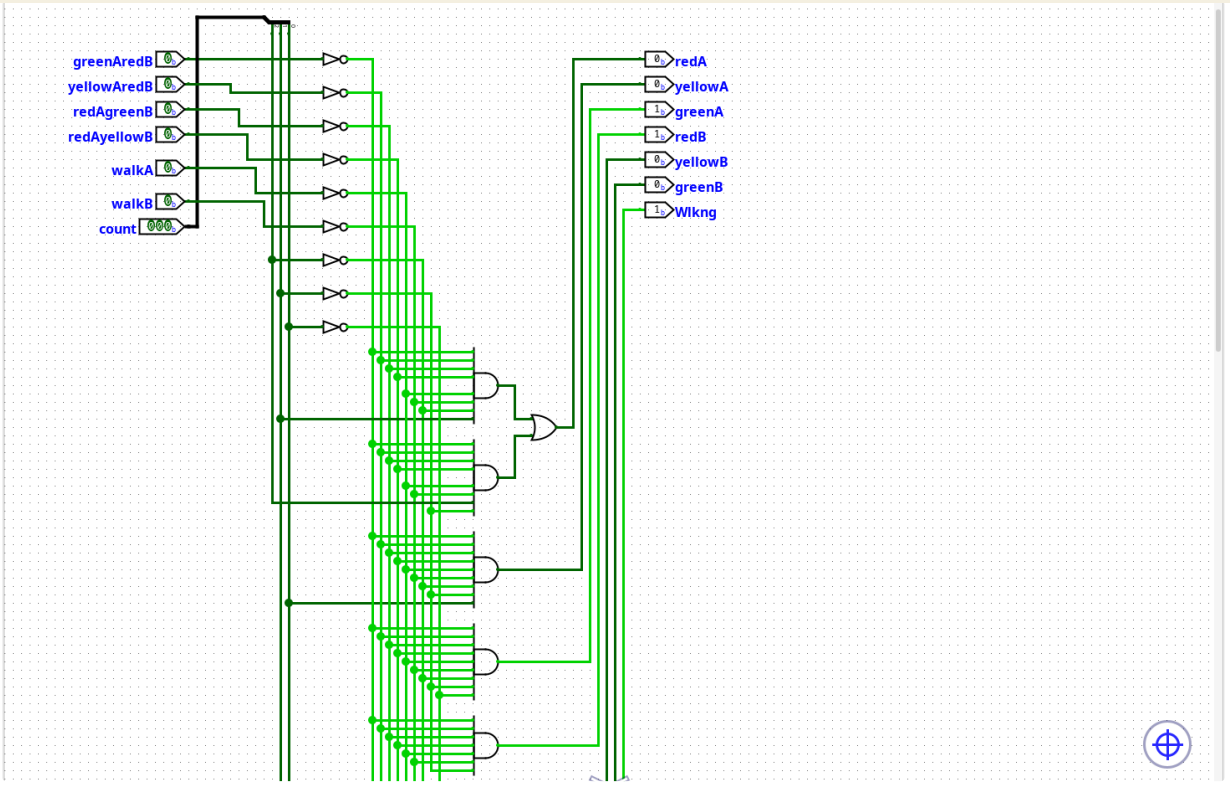
Next, I built a logic table to connect from the FSM to the lights, using Logisim to create a circuit in the process. Looking back on the project while writing the lab information, I assume this is where I broke the circuit.

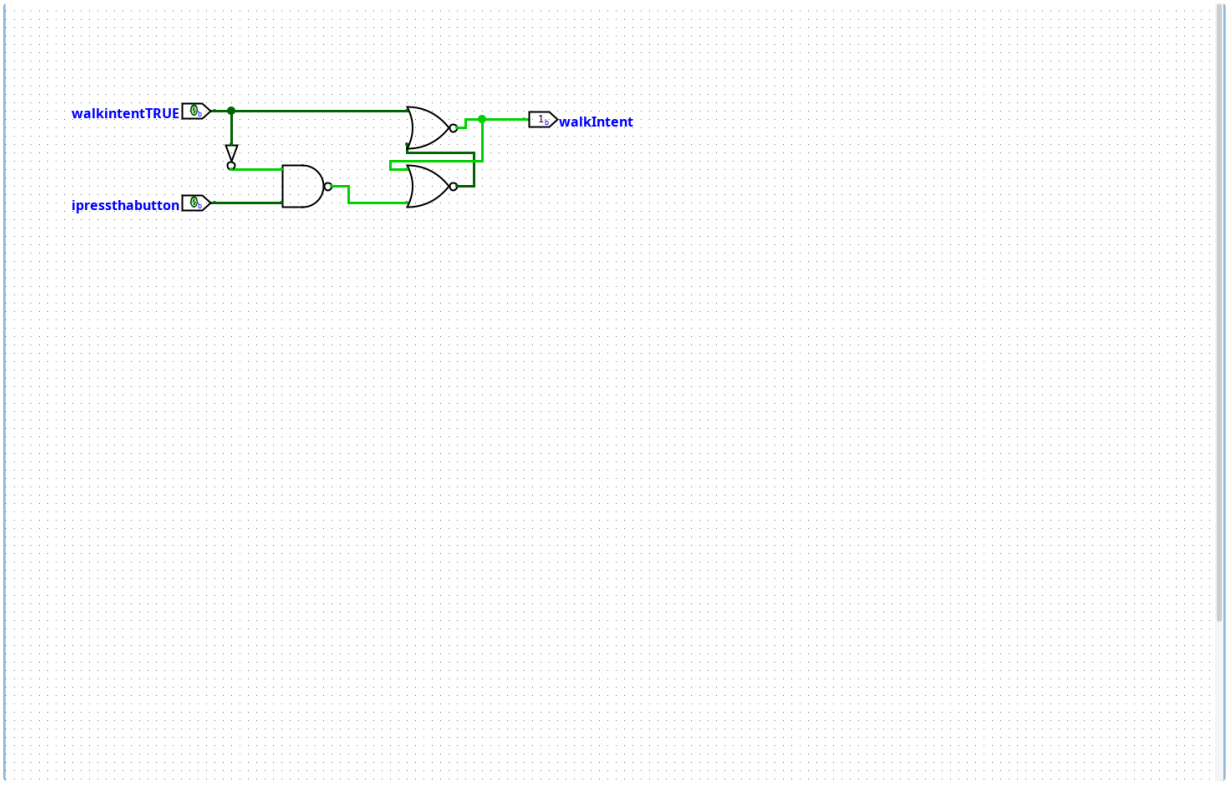
Decode	redA	yellowA	greenA	redB	yellowB	greenB
greenAredB	0	0	1	1	0	0
yellowAredB	0	1	0	1	0	0
redAgreenB	1	0	0	0	0	1
redAyellowB	1	0	0	0	1	0
walkA	1	0	0	1	0	0
walkB	1	0	0	1	0	0

Screenshot of blocking:

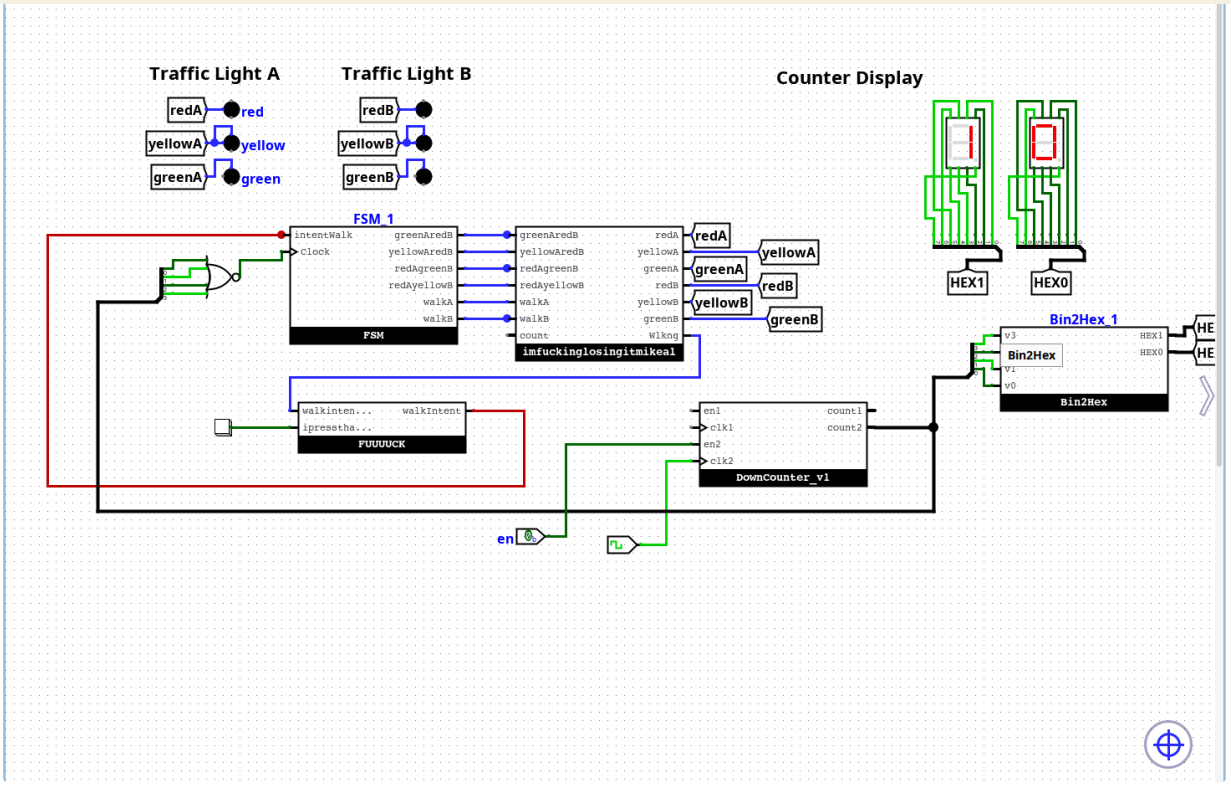








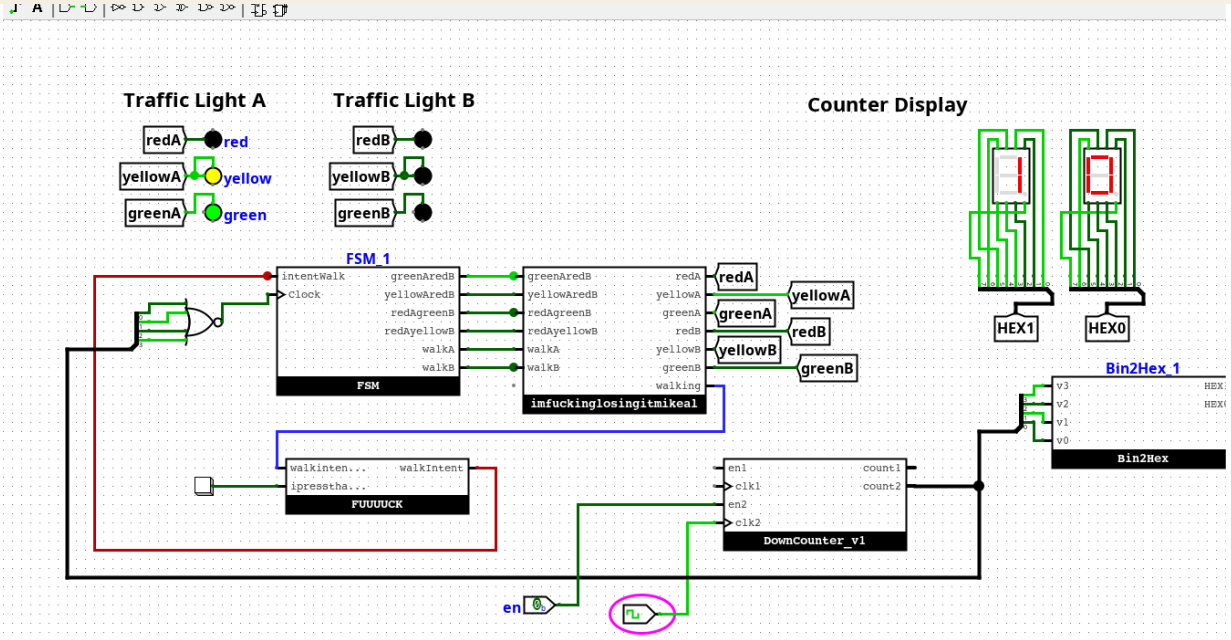
And this is where I wound up. The lights currently do not work, the walkIntent is red, but the counter works, and its fully able to switch from 3 second intervals to 10 second intervals on walkIntent.

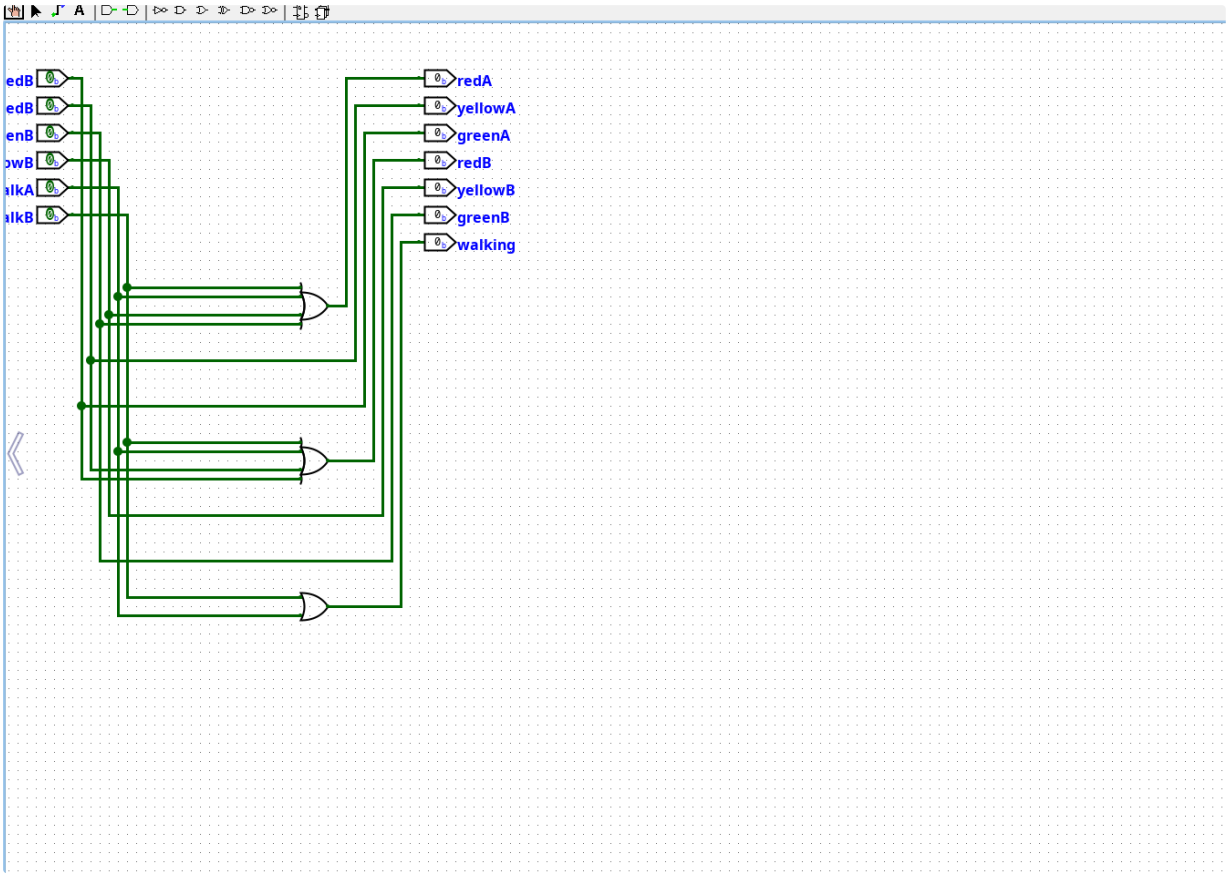


So I was like, okay, what did I do wrong?

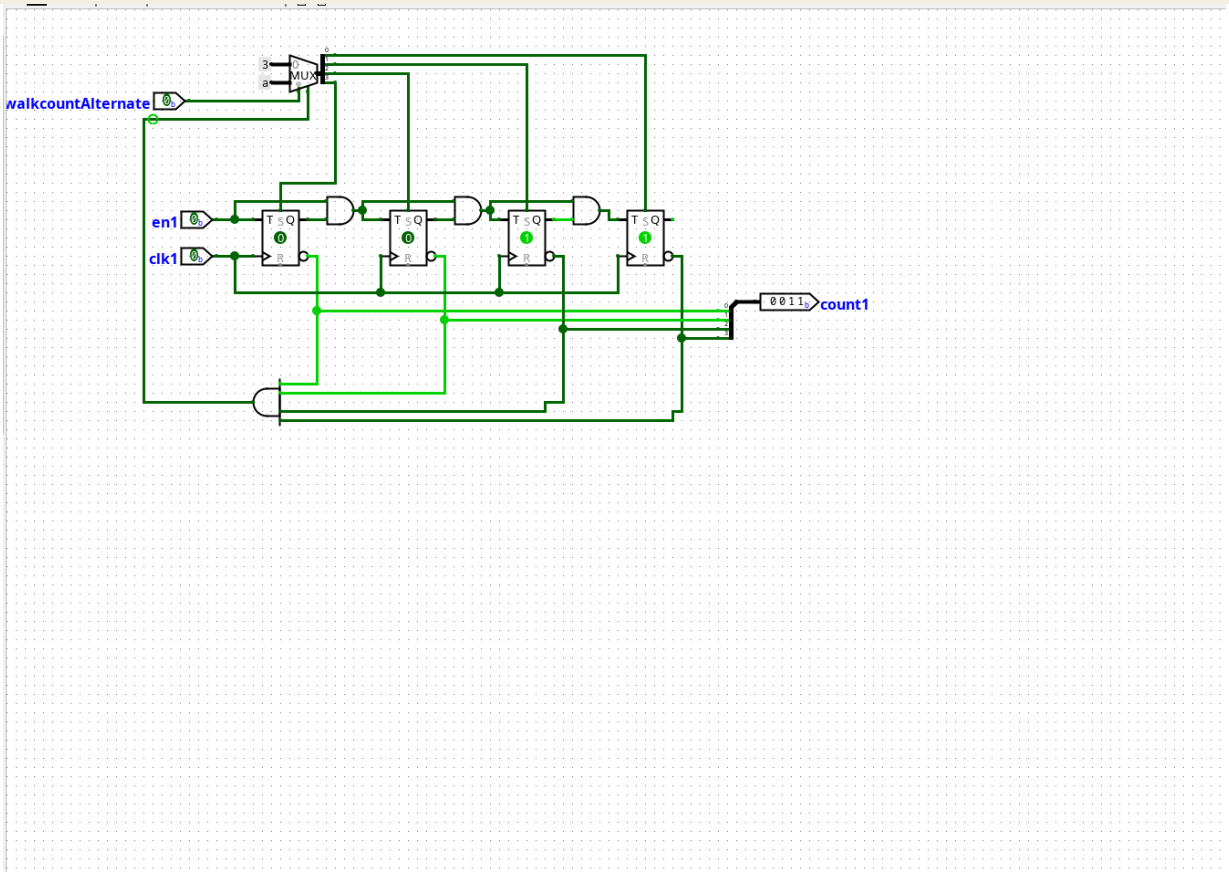
First off, I looked back at the truth table generated circuit. It doesn't work at all, not for what I want. I don't know if I typed it wrong, I have no idea what was wrong, so I used an LLM to help me manually make a circuit.

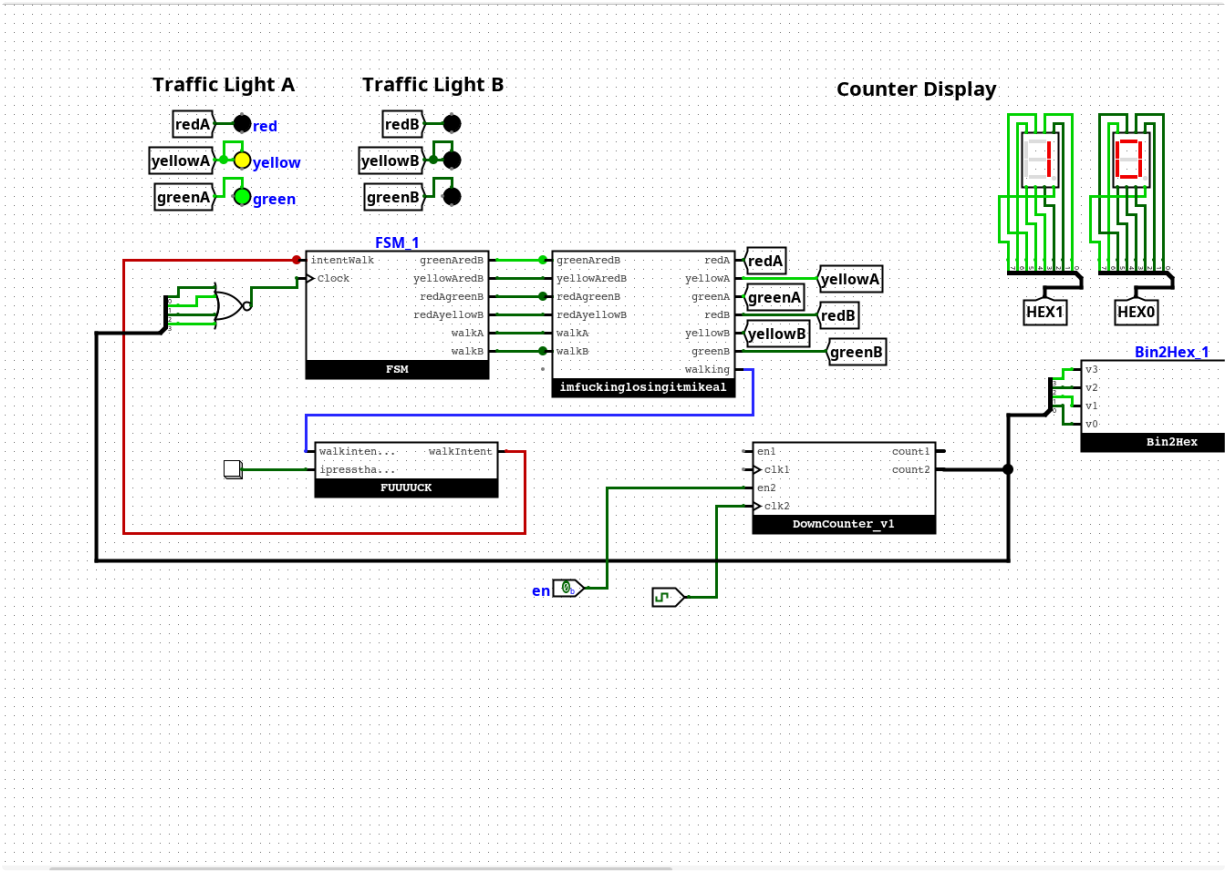
It didn't help at all.





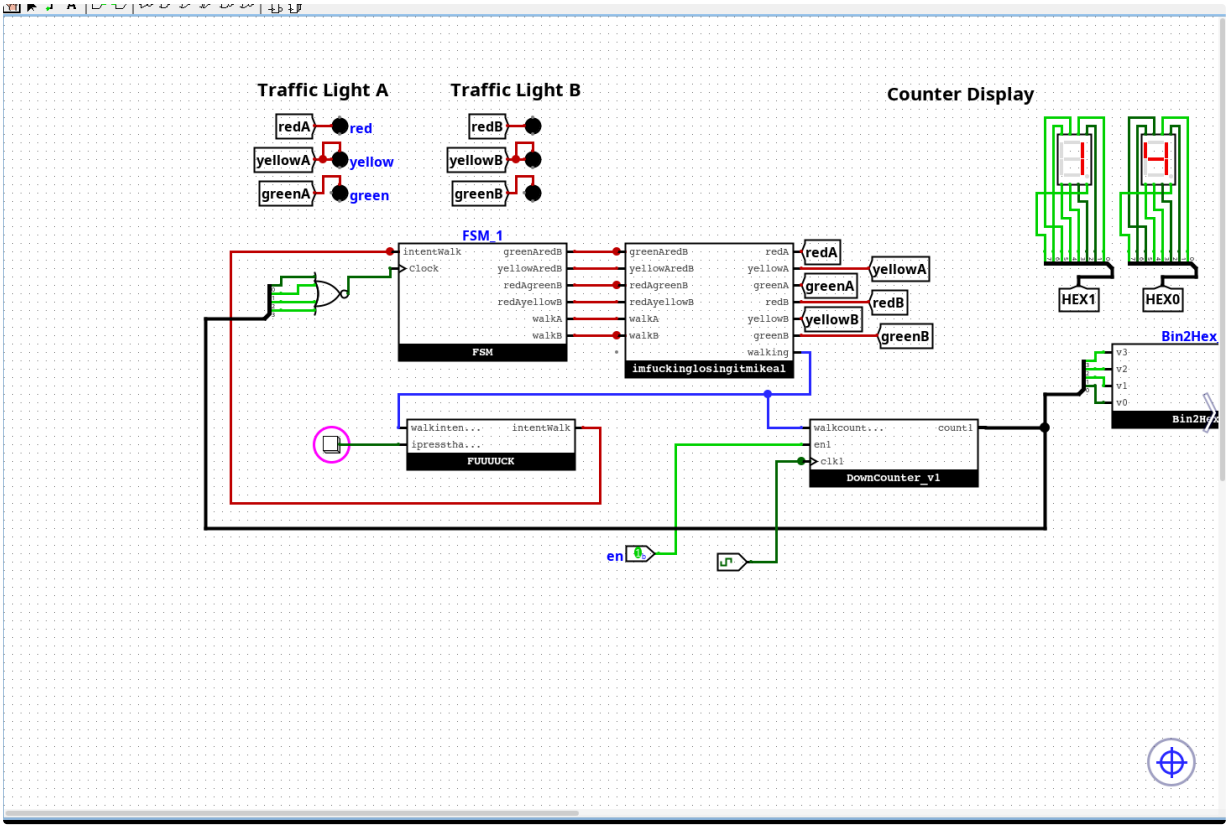
I ended up fixing it manually using equations instead of a truth table, but the problem still resides in the latch for walk I made.





Found this error: **Oscillation apparent**

At this point, I've put 6 hours of research into the project, and I called it there, kind of pulling my hair out.



What the project currently does: Starts counting from 10, goes green, yellow, red on light A. Once it hits red, it starts counting down from 15, then all the lights go dark, as displayed above. I'm unsure why my latch keeps breaking, and I would love any advice as to how I can fix this. I assume it has to do with the latch, but I can't say for certain.